

cosTapered

Model and Software Details

Andrew O. Finley, Lu Zhang, Arne Nothdurft, and Sudipto Banerjee

Updated June 18, 2026

Bayesian change-of-support modeling with tapered exponential covariance

Contents

1 Purpose	1
2 Spatial supports and design matrix	1
3 Fine-support Gaussian model	1
4 Observed-support model	2
5 Priors and nugget reparameterization	3
6 Collapsed likelihood and Metropolis updates	3
7 Recovery of regression and observed-support effects	4
8 Binomial logit response model	4
8.1 Polya-Gamma augmentation	5
8.2 Covariance updates in the binomial path	5
8.3 Binomial prediction	6
9 Fixed-size negative-binomial response model	6
10 Prediction targets and uncertainty	7
11 Cross-validation and computational guidance	7
11.1 Practical prediction guidance	8
12 Missing raster cells and support targets	10
13 Path for future response models	10

1 Purpose

This document records the model and computing details behind the current `costapered` implementation. It is intended as a compact mathematical reference for package users and developers. The introductory and model vignettes remain the best place to learn the user workflow; this document collects notation, model equations, posterior calculations, and software conventions in one place.

The package is intentionally narrower than the broader change-of-support framework that motivated it. The Gaussian response model follows the tapered exponential residual spatial process used in the forest damage application of [Zhang et al. \[2024\]](#). The current development version also includes binomial logit and fixed-size negative-binomial response models fit with Polya-Gamma data augmentation [[Polson et al., 2013](#), [Windle, 2013](#)]. The support-weight layer is shared by these response models; what changes is the observation model and the associated sampling step.

2 Spatial supports and design matrix

Let A_i , $i = 1, \dots, n_A$, denote fine raster cells. Let s_i denote the centroid of cell A_i . Let B_l , $l = 1, \dots, n_B$, denote observed supports, such as field plots, and let U_k , $k = 1, \dots, n_U$, denote prediction supports, such as stands or holdout polygons.

The fine-support covariate vector is $\mathbf{x}(A_i) \in \mathbb{R}^p$. The fine-support design matrix is

$$\mathbf{X} = \begin{bmatrix} \mathbf{x}(A_1)^\top \\ \vdots \\ \mathbf{x}(A_{n_A})^\top \end{bmatrix}. \quad (1)$$

The package uses the columns supplied in `X_rast`. It does not add an intercept, transform covariates, or scale raster layers automatically. If an intercept is wanted, the user supplies an intercept raster layer.

Observed-support weights are

$$h_{li} = \frac{|B_l \cap A_i|}{|B_l|}, \quad (2)$$

and define the sparse aggregation matrix $\mathbf{H}_{BA}$. Prediction-support weights are

$$g_{ki} = \frac{|U_k \cap A_i|}{|U_k|}, \quad (3)$$

and define the conceptual aggregation matrix $\mathbf{H}_{UA}$. The package stores $\mathbf{H}_{BA}$ as a sparse matrix and stores prediction weights block by block.

3 Fine-support Gaussian model

The current Gaussian response model is written at the finite raster-cell support as

$$y(A_i) = \eta(A_i) + \epsilon(A_i), \quad \eta(A_i) = \mathbf{x}(A_i)^\top \boldsymbol{\beta} + \omega(A_i). \quad (4)$$

In vector form,

$$\mathbf{y}_A = \boldsymbol{\eta}_A + \boldsymbol{\epsilon}_A, \quad \boldsymbol{\eta}_A = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\omega}_A, \quad (5)$$

with

$$\boldsymbol{\epsilon}_A \mid \tau^2 \sim N(\mathbf{0}, \tau^2 \mathbf{I}_{n_A}). \quad (6)$$

The independent noise term is a finite-cell nugget. It is not modeled as a continuous point-level white-noise process. This convention determines how nugget variance changes under support averaging.

The residual spatial process is

$$\boldsymbol{\omega}_A \mid \sigma^2, \phi, \gamma \sim N\{\mathbf{0}, \sigma^2 \mathbf{C}_A^{tap}(\phi, \gamma)\}. \quad (7)$$

For cells i and j ,

$$[\mathbf{C}_A^{tap}(\phi, \gamma)]_{ij} = \exp\{-\phi d(s_i, s_j)\} T_\gamma\{d(s_i, s_j)\}, \quad (8)$$

where $d(\cdot, \cdot)$ is Euclidean distance in the projected coordinate system. The current package supports the Wendland taper and the spherical taper. For $r = d/\gamma$, the Wendland taper used by `taper_code = 1L` is

$$T_\gamma(d) = (1 - r)^4(1 + 4r) \mathbf{1}(0 \leq r < 1), \quad (9)$$

and the spherical taper used by `taper_code = 2L` is

$$T_\gamma(d) = \left(1 - \frac{3}{2}r + \frac{1}{2}r^3\right) \mathbf{1}(0 \leq r < 1). \quad (10)$$

Both tapers are zero for distances greater than or equal to γ . The exponential effective range reported by the package is the distance at which the exponential component alone drops to 0.05:

$$r_{0.05} = -\log(0.05)/\phi \approx 3/\phi. \quad (11)$$

This effective range is not the taper distance γ . The taper distance is a compact-support cutoff chosen by the user.

4 Observed-support model

Observed responses are support averages of the fine-support response:

$$y(B_l) \approx \sum_i h_{li} y(A_i). \quad (12)$$

Stacking observed responses gives

$$\mathbf{y}_B = \mathbf{H}_{BA} \mathbf{X} \boldsymbol{\beta} + \mathbf{H}_{BA} \boldsymbol{\omega}_A + \mathbf{H}_{BA} \boldsymbol{\epsilon}_A. \quad (13)$$

Define

$$\mathbf{X}_B = \mathbf{H}_{BA} \mathbf{X}, \quad \boldsymbol{\omega}_B = \mathbf{H}_{BA} \boldsymbol{\omega}_A, \quad \mathbf{D}_h = \mathbf{H}_{BA} \mathbf{H}_{BA}^\top. \quad (14)$$

Then

$$\mathbf{y}_B \mid \boldsymbol{\beta}, \boldsymbol{\omega}_B, \tau^2 \sim N(\mathbf{X}_B \boldsymbol{\beta} + \boldsymbol{\omega}_B, \tau^2 \mathbf{D}_h), \quad (15)$$

and

$$\boldsymbol{\omega}_B \mid \sigma^2, \phi, \gamma \sim N\{\mathbf{0}, \sigma^2 \mathbf{C}_B^{tap}(\phi, \gamma)\}, \quad \mathbf{C}_B^{tap} = \mathbf{H}_{BA} \mathbf{C}_A^{tap} \mathbf{H}_{BA}^\top. \quad (16)$$

If observed supports do not overlap in raster cells, $\mathbf{D}_h$ is diagonal. If observed supports overlap, $\mathbf{D}_h$ can have off-diagonal entries. The implementation uses the general matrix expression.

5 Priors and nugget reparameterization

The current implementation uses a proper normal regression prior,

$$\boldsymbol{\beta} \sim N(\boldsymbol{\mu}_\beta, \mathbf{V}_\beta), \quad (17)$$

inverse-gamma variance priors in shape-scale form,

$$\tau_B^2 \sim \text{IG}(a_\tau, b_\tau), \quad \sigma^2 \sim \text{IG}(a_\sigma, b_\sigma), \quad (18)$$

and a bounded uniform prior for the exponential decay parameter,

$$\phi \sim \text{Uniform}(\phi_L, \phi_U). \quad (19)$$

The model is written in terms of the fine-support nugget variance τ^2 , but the sampler and prior are placed on

$$\tau_B^2 = \tau^2 \bar{d}_h, \quad \bar{d}_h = \frac{1}{n_B} \sum_{l=1}^{n_B} \mathbf{D}_h(l, l). \quad (20)$$

Thus

$$\tau^2 = \tau_B^2 / \bar{d}_h. \quad (21)$$

This is a reparameterization, not a different model. It makes prior specification easier because τ_B^2 is the average observed-support nugget variance contribution. It does not imply that every observed support has nugget variance τ_B^2 .

6 Collapsed likelihood and Metropolis updates

The sampler updates covariance parameters from a collapsed posterior in which $\boldsymbol{\beta}$ and $\boldsymbol{\omega}_B$ have been integrated out. For spatial fits let

$$\boldsymbol{\theta}_B = (\tau_B^2, \sigma^2, \phi).$$

Given $\boldsymbol{\theta}_B$, the marginal model is

$$\mathbf{y}_B \mid \boldsymbol{\theta}_B \sim N(\mathbf{X}_B \boldsymbol{\mu}_\beta, \mathbf{V}_Y), \quad (22)$$

where

$$\mathbf{V}_Y = \mathbf{X}_B \mathbf{V}_\beta \mathbf{X}_B^\top + \sigma^2 \mathbf{C}_B^{\text{tap}}(\phi, \gamma) + \frac{\tau_B^2}{\bar{d}_h} \mathbf{D}_h. \quad (23)$$

For nonspatial fits, the spatial covariance term is omitted.

The transformed parameters for spatial fits are

$$z_\tau = \log \tau_B^2, \quad z_\sigma = \log \sigma^2, \quad z_\phi = \log \left(\frac{\phi - \phi_L}{\phi_U - \phi} \right). \quad (24)$$

The Metropolis target includes the inverse-gamma and uniform prior terms plus the log-Jacobian terms for this transformation. The package stores both natural-scale samples and unconstrained-scale samples.

7 Recovery of regression and observed-support effects

After fitting, `cos_recover_B()` selects retained covariance parameter draws and samples $\boldsymbol{\beta}$ and $\boldsymbol{\omega}_B$ from their Gaussian full conditional. Define

$$\boldsymbol{\alpha} = \begin{bmatrix} \boldsymbol{\beta} \\ \boldsymbol{\omega}_B \end{bmatrix}, \quad \mathbf{Z} = [\mathbf{X}_B, \mathbf{I}_{n_B}]. \quad (25)$$

For a retained draw of τ^2, σ^2, ϕ ,

$$\mathbf{y}_B \mid \boldsymbol{\alpha}, \tau^2 \sim N(\mathbf{Z}\boldsymbol{\alpha}, \tau^2 \mathbf{D}_h), \quad (26)$$

with block Gaussian prior

$$\boldsymbol{\alpha} \sim N \left[\begin{bmatrix} \boldsymbol{\mu}_\beta \\ \mathbf{0} \end{bmatrix}, \begin{bmatrix} \mathbf{V}_\beta & 0 \\ 0 & \sigma^2 \mathbf{C}_B^{tap} \end{bmatrix} \right]. \quad (27)$$

Let $\mathbf{D}_h^{-1}$ denote the inverse of the observed-support nugget scaling matrix. The Gaussian full conditional is

$$\boldsymbol{\alpha} \mid \mathbf{y}_B, \tau^2, \sigma^2, \phi \sim N(\boldsymbol{\mu}_\alpha, \mathbf{Q}_\alpha^{-1}), \quad (28)$$

with

$$\mathbf{Q}_\alpha = \frac{1}{\tau^2} \mathbf{Z}^\top \mathbf{D}_h^{-1} \mathbf{Z} + \begin{bmatrix} \mathbf{V}_\beta^{-1} & 0 \\ 0 & (\sigma^2 \mathbf{C}_B^{tap})^{-1} \end{bmatrix}, \quad (29)$$

and

$$\boldsymbol{\mu}_\alpha = \mathbf{Q}_\alpha^{-1} \left\{ \frac{1}{\tau^2} \mathbf{Z}^\top \mathbf{D}_h^{-1} \mathbf{y}_B + \begin{bmatrix} \mathbf{V}_\beta^{-1} \boldsymbol{\mu}_\beta \\ \mathbf{0} \end{bmatrix} \right\}. \quad (30)$$

The implementation samples this conditional using a Cholesky factorization of $\mathbf{Q}_\alpha$. For nonspatial fits, $\boldsymbol{\omega}_B = \mathbf{0}$ and only $\boldsymbol{\beta}$ is recovered.

8 Binomial logit response model

For binary or binomial observed-support responses, let z_l be the number of successes out of m_l trials on observed support B_l . The current binomial model keeps the same observed-support linear predictor

$$\eta_l = \mathbf{x}_B(B_l)^\top \boldsymbol{\beta} + \omega_B(B_l), \quad (31)$$

but changes the observation distribution to

$$z_l \mid \eta_l, m_l \sim \text{Binomial}\{m_l, \pi_l\}, \quad \pi_l = \frac{\exp(\eta_l)}{1 + \exp(\eta_l)}. \quad (32)$$

Here $\mathbf{x}_B(B_l)^\top$ is row l of $\mathbf{X}_B = \mathbf{H}_{BA}\mathbf{X}$. The latent spatial effect $\boldsymbol{\omega}_B$ keeps the same tapered Gaussian prior,

$$\boldsymbol{\omega}_B \mid \sigma^2, \phi, \gamma \sim N\{\mathbf{0}, \sigma^2 \mathbf{C}_B^{tap}(\phi, \gamma)\}. \quad (33)$$

There is no Gaussian nugget variance τ^2 in this binomial likelihood. Extra-binomial variation, if needed, should be handled by a separate future model rather than by reusing the Gaussian finite-cell nugget.

8.1 Polya-Gamma augmentation

The binomial logit likelihood can be written using Polya-Gamma latent variables ξ_l [Polson et al., 2013]. Define

$$\kappa_l = z_l - m_l/2, \quad \xi_l \mid \eta_l, m_l \sim \text{PG}(m_l, \eta_l). \quad (34)$$

Conditional on ξ_l , the likelihood contribution is proportional to

$$\exp \left\{ \kappa_l \eta_l - \frac{1}{2} \xi_l \eta_l^2 \right\}, \quad (35)$$

which is Gaussian in the linear predictor. Stacking all observed supports, let

$$\boldsymbol{\kappa} = \mathbf{z} - \mathbf{m}/2, \quad \boldsymbol{\Omega} = \text{diag}(\xi_1, \dots, \xi_{n_B}), \quad \mathbf{Z} = [\mathbf{X}_B, \mathbf{I}_{n_B}],$$

and let $\boldsymbol{\alpha} = (\boldsymbol{\beta}^\top, \boldsymbol{\omega}_B^\top)^\top$. Conditional on $\boldsymbol{\Omega}$, the augmented likelihood is proportional to

$$\exp \left\{ \boldsymbol{\kappa}^\top \mathbf{Z} \boldsymbol{\alpha} - \frac{1}{2} \boldsymbol{\alpha}^\top \mathbf{Z}^\top \boldsymbol{\Omega} \mathbf{Z} \boldsymbol{\alpha} \right\}. \quad (36)$$

Combining this likelihood with the normal prior for $\boldsymbol{\beta}$ and the tapered Gaussian prior for $\boldsymbol{\omega}_B$ gives a Gaussian full conditional

$$\boldsymbol{\alpha} \mid \boldsymbol{\Omega}, \sigma^2, \phi, \mathbf{z} \sim N(\boldsymbol{\mu}_\alpha, \mathbf{Q}_\alpha^{-1}), \quad (37)$$

where

$$\mathbf{Q}_\alpha = \mathbf{Z}^\top \boldsymbol{\Omega} \mathbf{Z} + \begin{bmatrix} \mathbf{V}_\beta^{-1} & 0 \\ 0 & (\sigma^2 \mathbf{C}_B^{\text{tap}})^{-1} \end{bmatrix} \quad (38)$$

and

$$\boldsymbol{\mu}_\alpha = \mathbf{Q}_\alpha^{-1} \left\{ \mathbf{Z}^\top \boldsymbol{\kappa} + \begin{bmatrix} \mathbf{V}_\beta^{-1} \boldsymbol{\mu}_\beta \\ \mathbf{0} \end{bmatrix} \right\}. \quad (39)$$

The package draws the Polya-Gamma variables using the `BayesLogit` random-number generator. This is the reason the binomial path keeps $\boldsymbol{\beta}$ and $\boldsymbol{\omega}_B$ explicit in the Markov chain: the augmentation depends on the current linear predictor $\boldsymbol{\eta}_B$. Unlike the Gaussian response path, the current binomial sampler does not collapse over $\boldsymbol{\beta}$ and $\boldsymbol{\omega}_B$.

8.2 Covariance updates in the binomial path

Given $\boldsymbol{\omega}_B$, the covariance parameters (σ^2, ϕ) are updated from their prior and the Gaussian prior density of $\boldsymbol{\omega}_B$:

$$\boldsymbol{\omega}_B \mid \sigma^2, \phi, \gamma \sim N\{\mathbf{0}, \sigma^2 \mathbf{C}_B^{\text{tap}}(\phi, \gamma)\}. \quad (40)$$

The current implementation uses the same tapered covariance construction and bounded ϕ parameterization as the Gaussian model, but the target is not the Gaussian collapsed likelihood for $\mathbf{y}_B$. This separation is important: the binomial data contribution enters through the Polya-Gamma augmented linear predictor update, while (σ^2, ϕ) are informed by the current draw of $\boldsymbol{\omega}_B$.

8.3 Binomial prediction

For a fine prediction location or areal support P , the package first forms the conditional Gaussian distribution for the residual spatial effect,

$$\omega_P \mid \omega_B, \sigma^2, \phi, \gamma. \quad (41)$$

It then computes

$$\eta_P = \mathbf{x}_P^\top \boldsymbol{\beta} + \omega_P, \quad \pi_P = \frac{\exp(\eta_P)}{1 + \exp(\eta_P)}. \quad (42)$$

With `spatial_uncertainty = "conditional_mean"`, ω_P is set to its conditional mean. With `spatial_uncertainty = "marginal"`, the package draws independent marginal conditional values for each prediction cell or areal unit and transforms each draw through the inverse-logit link. These probability summaries are posterior summaries of π_P , not Gaussian delta-method approximations.

The current binomial prediction path does not produce joint fine-grid or joint areal probability samples. This is intentional for the present package stage: independent marginal prediction is sufficient for cell-wise and unit-wise maps and avoids dense prediction covariance matrices. Joint prediction can be added later if users need posterior summaries of spatial contrasts or totals on the probability scale.

9 Fixed-size negative-binomial response model

For observed-support count responses, the current package includes a fixed-size negative-binomial model. Let y_l be the count on observed support B_l , let $r > 0$ be the fixed `size` argument, and let o_l be an optional known offset, such as log exposure. The model uses the mean parameterization

$$y_l \mid \mu_l, r \sim \text{NB}(r, \mu_l), \quad E(y_l) = \mu_l, \quad \text{var}(y_l) = \mu_l + \mu_l^2/r, \quad (43)$$

with

$$\eta_l = \log \mu_l = o_l + \mathbf{x}_B(B_l)^\top \boldsymbol{\beta} + \omega_B(B_l). \quad (44)$$

There is no Gaussian nugget variance τ^2 in this likelihood. The count variability is defined by the negative-binomial likelihood and the fixed size r . The package does not estimate r in the current implementation; users should choose it from prior information or examine a sensitivity grid.

The Polya-Gamma augmentation uses the negative-binomial logit-probability tilt

$$\psi_l = \eta_l - \log r, \quad (45)$$

and draws

$$\xi_l \mid y_l, r, \psi_l \sim \text{PG}(y_l + r, \psi_l). \quad (46)$$

Conditional on ξ_l , the likelihood is Gaussian in the linear predictor. The corresponding working response for $\mathbf{x}_B(B_l)^\top \boldsymbol{\beta} + \omega_B(B_l)$ is

$$z_l = \frac{(y_l - r)/2}{\xi_l} + \log r - o_l. \quad (47)$$

The Gaussian update for $\boldsymbol{\alpha} = (\boldsymbol{\beta}^\top, \boldsymbol{\omega}_B^\top)^\top$ then has the same form as the binomial Polya-Gamma update, with $\boldsymbol{\Omega} = \text{diag}(\xi_l)$ and $\mathbf{z}$ replacing the binomial working response.

For prediction, the package reports link-scale summaries for η_P and mean-count summaries for $\mu_P = \exp(\eta_P)$. If `target = "observed"`, it also simulates count draws from $\text{NB}(r, \mu_P)$. Prediction offsets must be supplied by the user; the package does not infer count exposure from polygon area.

10 Prediction targets and uncertainty

The latent fine-support target is

$$\boldsymbol{\eta}_P = \mathbf{X}_P \boldsymbol{\beta} + \boldsymbol{\omega}_P. \quad (48)$$

For spatial fits,

$$\boldsymbol{\omega}_P \mid \boldsymbol{\omega}_B, \sigma^2, \phi, \gamma \sim N\{\mathbf{C}_{PB} \mathbf{C}_B^{-1} \boldsymbol{\omega}_B, \sigma^2(\mathbf{C}_{PP} - \mathbf{C}_{PB} \mathbf{C}_B^{-1} \mathbf{C}_{BP})\}. \quad (49)$$

In `cos_predict_fine()`, `conditional_mean` uses the conditional mean of $\boldsymbol{\omega}_P$, so uncertainty summaries omit conditional spatial prediction uncertainty. The `marginal` option draws independent marginal conditional values for each fine prediction location, and `joint` draws from the joint conditional distribution across the fine prediction locations.

For an areal prediction support U_k , the latent target is

$$\eta(U_k) = \sum_i g_{ki} \eta(A_i). \quad (50)$$

If the requested target is an observed response, the package adds finite-cell nugget variation:

$$y(U_k) = \eta(U_k) + \epsilon(U_k), \quad \epsilon(U_k) \sim N\left(0, \tau^2 \sum_i g_{ki}^2\right). \quad (51)$$

Current areal prediction samples are marginal by prediction unit. They include one-dimensional conditional uncertainty for each U_k , but they are not joint samples across all prediction units. They should not be used for totals, contrasts, rankings, or other summaries that require cross-unit posterior covariance. The package reports areal means; users compute totals after choosing area units.

11 Cross-validation and computational guidance

`cos_cv_observed()` refits the model across observed-support folds and scores predictions on held-out supports. For Gaussian fits, the validation target should match the inferential question. The latent target compares predictions of $\eta(B_l)$, while the observed target compares predictions of $y(B_l) = \eta(B_l) + \epsilon(B_l)$ and includes the support-averaged nugget. If a column containing latent validation values is available, both targets can be scored from the same fold refits. Otherwise observed-response validation is the natural default.

The package reports RMSPE, MAE, bias, correlation, slope, CRPS, 95% interval coverage, mean 95% interval width, and mean error. Coverage should be read together with interval width: a model can cover held-out observations by using wide predictive intervals without providing useful sharp predictions. CRPS is often a useful complement because it rewards predictive distributions that are both calibrated and concentrated.

Computationally, the dominant costs are covariance construction, Cholesky factorizations, and prediction covariance calculations. Tapering makes observed-support covariance construction sparse before aggregation, while the collapsed Gaussian sampler avoids sampling $\boldsymbol{\beta}$ and $\boldsymbol{\omega}_B$ inside the Metropolis update. Fine-grid joint prediction is the expensive path because it forms dense prediction covariance matrices. Marginal prediction avoids the joint dense matrix and is usually the appropriate choice for cell-wise or per-polygon posterior means, standard deviations, and intervals. The `conditional_mean` option is fastest but omits conditional spatial prediction uncertainty.

11.1 Practical prediction guidance

The prediction functions expose three separate decisions: the prediction support, the response target, and how to handle conditional spatial uncertainty. These choices determine which downstream summaries are coherent. The first question is the inferential target, not the runtime. For Gaussian fits, `target = "latent"` estimates the underlying support mean, excluding finite-cell or support-averaged nugget variation. This is usually the target for mapped condition, stand means, polygon means, and inventory summaries. `target = "observed"` adds the Gaussian nugget contribution on the requested support, so it targets the support response with nugget variation included. For individual stands or polygons, `spatial_uncertainty = "marginal"` is the statistically coherent choice for posterior means, standard deviations, and credible intervals because it includes the one-dimensional conditional uncertainty for each prediction unit. Those draws are independent across prediction units, so they are not coherent for totals, rankings, contrasts, or allocation decisions that depend on cross-unit posterior covariance. `spatial_uncertainty = "conditional_mean"` is a plug-in summary of the conditional mean spatial surface; it can be useful for fitted surfaces and posterior mean displays, but it omits residual conditional prediction uncertainty and gives intervals that are too narrow if interpreted as full prediction uncertainty. Computational expense should be considered after the estimand is chosen, not as the rule for choosing the estimand.

Figure 1 summarizes these decisions. Box color identifies what kind of prediction sample is being represented: plug-in conditional means, independent marginal prediction-unit samples, or joint fine-support Gaussian samples.

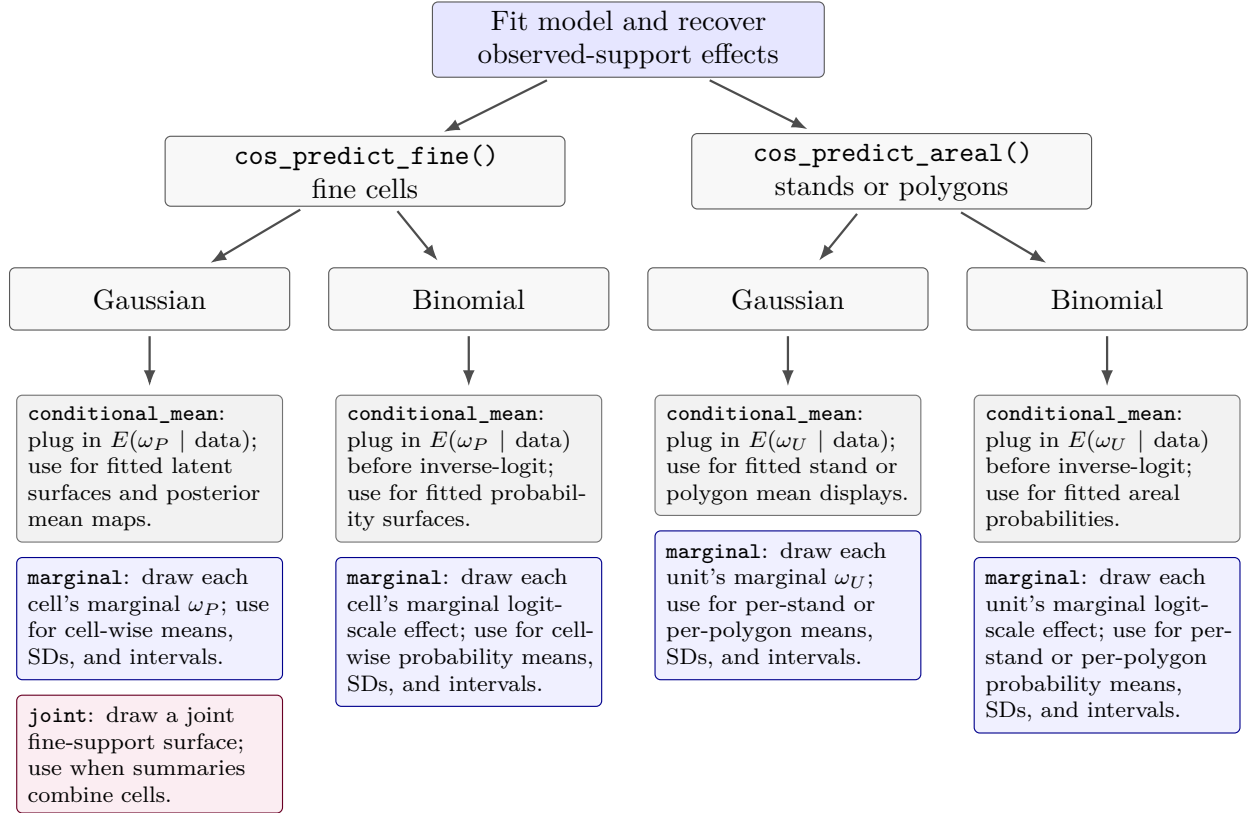


Figure 1: Prediction choices in `cosTapered`. The key distinction is whether samples represent conditional means, independent marginal uncertainty for each prediction unit, or a joint fine-support Gaussian surface. Gray boxes are plug-in conditional-mean summaries. Blue boxes are marginal prediction-unit summaries, appropriate for stand, polygon, or cell estimates interpreted one unit at a time. Purple boxes are joint fine-support samples for summaries that combine cells. Runtime can differ substantially among paths, but the inferential target should determine the choice.

12 Missing raster cells and support targets

Support weights define the estimand. If raster covariates are incomplete over part of an observed or prediction polygon, dropping those cells changes the support average. The default package behavior is therefore `missing = "error"` when incomplete covariates would leave a support row sum materially below one. Two explicit alternatives are available:

- `missing = "drop"` keeps the covered-cell weights without rescaling, so row sums may be below one.
- `missing = "renormalize"` rescales the retained weights to sum to one, changing the target to the covered-cell average.

These choices should be made as modeling decisions, not as preprocessing accidents.

13 Path for future response models

The support-weight machinery is largely independent of the response model. Future response models should reuse the same objects $\mathbf{H}_{BA}$, $\mathbf{H}_{UA}$, $\mathbf{X}_B$, and support-weight diagnostics where possible. The parts likely to change are:

- the observation distribution for $\mathbf{y}_B$;
- the latent linear predictor or link function;
- the sampler used for covariance and latent-effect updates;
- the posterior predictive distribution for observed targets.

The distinction between latent support averages and observed support responses should remain explicit for any new response family.

References

- Nicholas G. Polson, James G. Scott, and Jesse Windle. Bayesian inference for logistic models using Polya-Gamma latent variables. *Journal of the American Statistical Association*, 108(504): 1339–1349, 2013. doi: 10.1080/01621459.2013.829001.
- Jesse Windle. *Forecasting High-Dimensional, Time-Varying Variance-Covariance Matrices with High-Frequency Data and Sampling Polya-Gamma Random Variates for Posterior Distributions Derived from Logistic Likelihoods*. PhD thesis, University of Texas at Austin, 2013.
- Lu Zhang, Andrew O. Finley, Arne Nothdurft, and Sudipto Banerjee. Bayesian modeling of incompatible spatial data: A case study involving Post-Adrian storm forest damage assessment. *International Journal of Applied Earth Observation and Geoinformation*, 135:104224, 2024. doi: 10.1016/j.jag.2024.104224.